

Tianshuo Han

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Education

University of Chinese Academy of Sciences | **Ph.D. Candidate** Sep 2021 — Jan 2027
(Expected)

Cyberspace Security, Institute of Information Engineering | Direct Ph.D. Program

Research interest focused on **Linux kernel security**, **software security**, **AI for security**. My research were published at top-tier security conferences including USENIX Security, CCS, S&P.

China Agricultural University | **Undergraduate** Sep 2017 — Jun 2021

Biological Sciences, College of Life Sciences | Bachelor of Science

Outstanding academic performance with multiple academic scholarships. Awarded National Second Prize in the China Undergraduate Mathematical Contest in Modeling and Gold Medal in the iGEM (International Genetically Engineered Machine) Competition.

Publications

NetPanic: A Fuzzer-in-the-Middle Approach to Auditing the Linux Kernel Network Stack's Remote Attack Surface

Tianshuo Han, Zong Cao, Zhen Dong, Xiapu Luo, Zhenyu Song, and Jian Liu

47th IEEE S&P, 2026, CCF-A

- Conducted the first systematic fuzzing-based audit of the Linux kernel network stack's remote attack surface. Proposed the Fuzzer-in-the-Middle architecture, discovering 15 new remotely triggerable kernel vulnerabilities.
- Independently completed the overall architecture design and implementation. Led the paper writing effort.

CARDSHARK: Understanding and Stabilizing Linux Kernel Concurrency Bugs Against the Odds

Tianshuo Han, Xiaorui Gong, and Jian Liu

33rd USENIX Security Symposium, 2024, CCF-A

- Conducted the first in-depth analysis of the root causes of instability in triggering kernel race condition vulnerabilities, and proposed a general-purpose exploitation technique that significantly improves the success rate and stability of kernel race condition exploitation.
- Independently completed the mechanism analysis of kernel race vulnerabilities, exploitation technique development, and performance evaluation. Led the paper writing effort.

Reviving Discarded Vulnerabilities: Exploiting Previously Unexploitable Linux Kernel Bugs Through Control Metadata Fields

Hao Zhang, Jie Lu, Shaomin Chen, Tianshuo Han, Bolun Zhang, Jian Liu, Xiaorui Gong

32nd ACM CCS, 2025, CCF-A

- Proposed a novel exploitation technique based on manipulating critical metadata fields in kernel structures, effectively enhancing exploitation capabilities with weak vulnerability primitives and expanding the scope of exploitable kernel vulnerabilities.
- Deeply involved in defining the core thesis and responding to reviewer comments. Participated in evaluation experiment design and CodeQL code development.

Whole-genome sequence and comparative analysis of *Trichoderma asperellum* ND-1 reveal its unique enzymatic system for efficient biomass degradation

Fengzhen Zheng, Tianshuo Han, Abdul Basit, Junquan Liu, Ting Miao, and Wei Jiang

Catalysts 12(4), 2022, IF=4.16

- Revealed the unique lignocellulose degradation enzyme system of *Trichoderma asperellum* ND-1 through whole-genome sequencing and secretome analysis, providing a genomic theoretical foundation for efficient biomass conversion.
- Responsible for in-depth genomic data analysis and scientific data visualization.

Kernel Fuzzing of Distributed File Systems via Intra-Cluster Protocol Poisoning

[First Author], **Tianshuo Han**, et al. (2nd author)

ACM CCS, 2026 (Under Submission), CCF-A

- Proposed *intra-cluster protocol poisoning* as a novel kernel attack surface for distributed file systems, and built a fuzzer that interposes on inter-node traffic and mutates protocol messages in flight, guided by static-analysis-based contamination-scope quantification and protocol-state-synchronized progressive exploration.
- Discovered XX kernel vulnerabilities (heap overflows, use-after-free, null-pointer dereferences) across five distributed file systems (CephFS, SMB, NFS, 9P, etc.) running the latest Linux kernel, all triggerable by poisoned messages alone.

Key Projects

NetPanic — A Novel Fuzzing Framework for Kernel Network Stack Nov 2023 - Present

- Designed a novel architecture distinct from Syzkaller, specifically optimized for kernel network stack fuzzing.
- Built on the LibAFL framework with deep customization, reimplementing core components including mutation, execution, feedback, and scheduling.
- Independently completed the overall architecture design and implementation, totaling 40K+ lines of code (Rust + C).
- Deep understanding of fuzzer design principles with proficiency in LibAFL framework internals.

Linux Kernel Rootkit Development Dec 2023 - May 2024

- Independently designed and implemented a full-featured Linux kernel rootkit, covering self-hiding, port hiding, privilege escalation, file/process hiding, and persistence across reboots.
- Core design philosophy: **lightweight architecture** and **high stealth**, achieving complete functionality with minimal code while evading mainstream rootkit detection techniques.
- **Broad compatibility**: supports all Linux kernel versions from 2.6 to 6.3.

Automated `sprintf()` Vulnerability Discovery in the Linux Kernel Oct 2022 - Jan 2023

- Systematically discovered buffer overflow vulnerabilities in the Linux kernel caused by insufficient buffer length reservation in `sprintf()` function calls using static analysis.
- Implemented automated analysis tooling based on the **LLVM** framework, identifying 100+ potential overflow sites.

Shadow Brokers Leaked Hacking Tools — In-depth Reverse Engineering Jan 2022 - May 2022

- Conducted in-depth analysis of exploit tools including EARLYSHOVEL, ELECTRICSLIDE, EVENLESSON, and TELEX.
- Reverse-engineered backdoor tools NOPEN and STOICSURGEON, and developed corresponding detection tools.

Security Contributions

CVE-2025-38501 | Linux Kernel KSMDB Remote DoS Aug 2025

- Discovered a design flaw in the Linux kernel KSMDB module when handling abnormal TCP connections, allowing remote attackers to exhaust the connection pool via TCP handshakes, rendering the service unavailable.
- Completed root cause analysis, PoC development, and assisted with the fix.

CVE-2025-38089, CVE-2025-38051, CVE-2025-40210, CVE-2025-40212, CVE-2026-23220 | Linux Kernel Remote-Triggerable Vulnerabilities 2025 - 2026

- Discovered five remote-triggerable vulnerabilities in the Linux kernel, spanning the KSMDB, NFS, and SUNRPC subsystems.
- Conducted vulnerability mechanism analysis, PoC construction, and participated in the patching process.

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Sep 2022

- Fixed a warning triggered in the Linux kernel memory management subsystem when the `madvise()` system call with the `MADV_PAGEOUT` flag processes a non-LRU page.
- Participated in patch development, testing, and verification.

Teaching Experience

Teaching Assistant — Malware Discovery and Analysis

Sep 2023 - Feb 2024

School of Cyberspace Security, University of Chinese Academy of Sciences

- Served as teaching assistant for a graduate-level course covering dynamic and static program analysis techniques for malware detection and analysis on Windows systems.
- Responsible for Q&A sessions, lab assignment grading, and exam question preparation.
- Helped students master core malware analysis techniques and develop practical hands-on skills.

Technical Skills

- **Programming Languages:** Expert in **C**; proficient in **Rust**, Python, Shell, $\text{\LaTeX}$; familiar with C++, Golang, Matlab, R
- **Fuzzing:** Deep expertise in LibAFL framework; familiar with AFL++, Syzkaller, and other mainstream fuzzing tools
- **Professional Skills:** Experienced in Linux kernel exploitation techniques; proficient in binary reverse engineering and exploitation; skilled with Shell, Vim, Git, GDB, IDA, and other development and security analysis tools

Personal Qualities

- Deep theoretical foundation and extensive engineering experience in Linux kernel security, with strong technical innovation capabilities.
- Highly self-motivated with excellent execution, passionate about technology and innovation, with outstanding communication and teamwork skills.
- Healthy lifestyle with high energy levels, maintaining a consistent fitness routine for years.
- Near-native English proficiency, capable of fluently reading and writing technical documentation in English.